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Lab 10.3 Machine Learning 3 - Real-time Sound Recognition for Classification

Entire remote monitoring system

3.1 MTConnect for pump monitoring

In the prvious section, we performed utilizing the CNN model by TensorFlow Lite to predict the running state of the vacuum pump. Let's make an entire monitoring system as Figure 1. Depending on the project or system requirements, you skip the MTConnect part. However, considering the potential use of MTConnect to collect machine data in the future, design and create an MTConnect agent to collect and merge the running state prediction (Event) and sound level (Sample) of the vacuum pump.

The idea is that we know both the execution state (OFF/ON, Event) and the vacuum state (Vacuuming/Air leaking, Event) from the CNN model. Also, we calculate the sound level (Sample). The schematic and data items are shown in Figure 7.

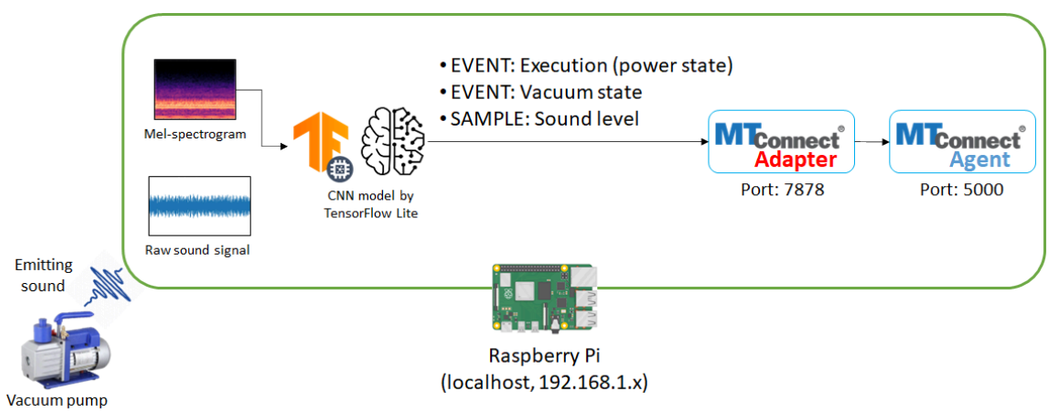


Figure 7 Schematic and data items of MTConnect

The examples of 'agent.cfg' and 'device.xml' are below. You may have different an XML data structure or data item definitions. Also, you can check the example of MTConnect agent running on the server computer: http://mepotrb16.ecn.purdue.edu:5000/. Recall the previous labs to run MTConnect service on Raspberry Pi and reuse the agent execution file and other directory/files (schema, style, and so on).

---

agent.cfg

```
Devices = device.xml
AllowPut = true
Port = 5000
ReconnectInterval = 1000
BufferSize = 17
SchemaVersion = 1.4
MonitorConfigFiles = true
Pretty = true
# MinimumConfigReloadAge = 30
```

```
Adapters {
   # Log file has all machines with device name prefixed
   Device {
      Device = vacuum_pump
      Host = 127.0.0.1
      Port = 7878
   }
}

Files {
    schemas {
        Path = ./schemas
        Location = /schemas/
    }
    styles {
        Path = ./styles
        Location = /styles/
    }
    Favicon {
        Path = ./styles/favicon.ico
        Location = /favicon.ico
    }
}

DevicesStyle { Location = /styles/Devices.xsl }
StreamsStyle { Location = /styles/Streams.xsl }

# Logger Configuration
logger_config
{
    logging_level = debug
}
```

---

device.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<MTConnectDevices xmlns:m="urn:mtconnect.org:MTConnectDevices:1.1" xmlns:xsi="http://www.w3.
    <Header creationTime="2010-03-04T18:44:40+00:00" sender="localhost" instanceId="12677282
    <Devices>
        <Device id="vacuum_pump" name="vacuum_pump" uuid="001">
            <DataItems>
                <DataItem category="EVENT" id="001" name="avail" type="AVAILABILITY"/>
                <DataItem category="EVENT" id="e1" name="execution" type="EXECUTION"/>
                <DataItem category="EVENT" id="vs1" name="vacuum_state" type="STATE"/>
                <DataItem category="SAMPLE" id="spl" name="sound_level" type="SOUND_LEVEL" u

            </DataItems>
        </Device>
    </Devices>
</MTConnectDevices>
```

The MTConnect adapter sample code (lab10_sample_adapter.py) is also given. The interpretation algorithm to take the CNN model output as MTConnect data item is shown in Figure 8. If the output

is 0, execution is "OFF" and vacuum state is "N/A" (not available). If the output is 1, execution is "ON" and vacuum state "Air leaking". Lastly, if the output is 2, execution is "ON" and vacuum state "Vacuuming". However, it is recommended to try programming the adapter yourself. If you use the given sample code, you have to modify some lines. The incomplete part of the sample script is below. You have to change the variables which is wrarpped by aserteric marks (*). By completing the sample code , perform Task 3.1.

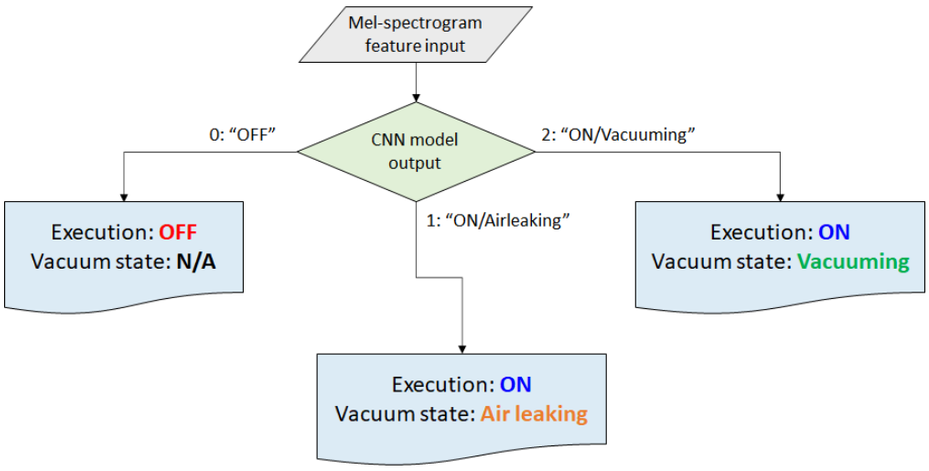


Figure 8 Algorithm to take MTConnect data item from CNN model output

Note that your CNN model (h5 format), mtconnect_adapter.py, and data_item.py must be in the same directory with the adapter.

Python - Python3 (Part of lab10_sample_adapter.py)

```
## == GLOBAL CONSTANT ==
SAMPLE = "sample" # sample string for MTConnect HTML sample method
CURRENT = "current" # current string for MTConnect HTML current method
SAMP_RATE = int(??) # sampling rate
*SAMP_RATE* = int(?) # sampling rate
*CHUNK* = int(?) # chunk size
*AGENT* = "http://?ip?:?port?/" # MTConnect agent ip:port

*N* = int(?) # number of sequence to take sound
*N_FFT* = int(?) # number of FFT
*N_MELS* = int(?) # number of Mel filter bank

*model_file* = '?20230309_212154_Prelab10_CNN_model1.h5?' # CNN model fileanme. Note that th
```

Task 3.1

Run MTConnect agent and adapter on Raspberry Pi with given informationa and sample. Capture '/current' response of the MTConnect agent as Figure 9 from **your laptop** and attach it to the report. While performing this, check if your adapter is working correctly. You see the terminal as Figure 10. **If nothing is shown in your browser, please close the "Pretty" style following the instruction of Task 2.1 in L4 Colab 2.**

Homepage - Spring 2026 ME | Home - Google Drive | index.ipynb - Colab | L10_Colab3.ipynb - Colab | MTConnect Device Streams | Ask Gemini

Not secure 192.168.1.4:5000/current

- creationTime: 2026-04-08T18:56:10Z
- sender: lamm-UltraPoint
- instanceId: 1775668676
- version: 1.8.0.2
- bufferSize: 131072
- nextSequence: 4348
- firstSequence: 1
- lastSequence: 4347

Device: vacuum_pump; UUID: 001

Device : vacuum_pump

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-04-08T18:56:09.876444Z	SoundLevel		sound_level	spl	4347	45.1514597705476

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-04-08T17:18:18.65902Z	Availability		avail	001	7	AVAILABLE
2026-04-08T18:53:49.074655Z	Execution		execution	e1	4247	OFF
2026-04-08T17:17:56.564273Z	AssetChanged			vacuum_pump_asset_chg	2	UNAVAILABLE
2026-04-08T17:17:56.562959Z	AssetRemoved			vacuum_pump_asset_rem	1	UNAVAILABLE
2026-04-08T18:53:49.074655Z	State		vacuum_state	vs1	4248	N/A

192.168.1.23 (WayVNC) - RealVNC Viewer

pi@AaryaFarheen: ~... adapter_sample adapter_sample pi@AaryaFarheen: ~

File Edit Tabs Help

```
2026-04-08T18:54:56.504815Z, Pump is now OFF and N/A state in 44.76 dB SPL.
2026-04-08T18:54:58.884947Z, Pump is now ON and Vacuuming state in 69.86 dB SPL.
2026-04-08T18:55:01.290109Z, Pump is now ON and Vacuuming state in 70.48 dB SPL.
2026-04-08T18:55:03.717206Z, Pump is now ON and Vacuuming state in 70.09 dB SPL.
2026-04-08T18:55:06.104569Z, Pump is now ON and Vacuuming state in 70.12 dB SPL.
2026-04-08T18:55:08.422648Z, Pump is now ON and Vacuuming state in 70.29 dB SPL.
2026-04-08T18:55:10.822123Z, Pump is now ON and Vacuuming state in 70.37 dB SPL.
2026-04-08T18:55:13.184833Z, Pump is now ON and Vacuuming state in 70.25 dB SPL.
2026-04-08T18:55:15.637749Z, Pump is now ON and Air leaking state in 70.52 dB SPL.
2026-04-08T18:55:17.964641Z, Pump is now ON and Air leaking state in 70.87 dB SPL.
2026-04-08T18:55:20.358544Z, Pump is now OFF and N/A state in 67.27 dB SPL.
2026-04-08T18:55:22.735264Z, Pump is now OFF and N/A state in 44.5 dB SPL.
2026-04-08T18:55:25.114429Z, Pump is now OFF and N/A state in 43.96 dB SPL.
2026-04-08T18:55:27.494175Z, Pump is now OFF and N/A state in 44.88 dB SPL.
2026-04-08T18:55:29.890031Z, Pump is now ON and Vacuuming state in 70.04 dB SPL.
2026-04-08T18:55:32.256185Z, Pump is now ON and Vacuuming state in 70.37 dB SPL.
2026-04-08T18:55:34.588035Z, Pump is now ON and Vacuuming state in 70.37 dB SPL.
2026-04-08T18:55:36.979594Z, Pump is now ON and Vacuuming state in 70.33 dB SPL.
2026-04-08T18:55:39.395740Z, Pump is now ON and Vacuuming state in 70.4 dB SPL.
2026-04-08T18:55:41.801229Z, Pump is now ON and Vacuuming state in 70.35 dB SPL.
2026-04-08T18:55:44.192948Z, Pump is now ON and Vacuuming state in 69.94 dB SPL.
2026-04-08T18:55:46.632465Z, Pump is now ON and Air leaking state in 70.51 dB SPL.
2026-04-08T18:55:49.036200Z, Pump is now ON and Air leaking state in 70.87 dB SPL.
2026-04-08T18:55:51.519250Z, Pump is now OFF and N/A state in 66.42 dB SPL.
2026-04-08T18:55:53.903225Z, Pump is now OFF and N/A state in 44.57 dB SPL.
2026-04-08T18:55:56.188060Z, Pump is now OFF and N/A state in 43.96 dB SPL.
2026-04-08T18:55:58.573576Z, Pump is now OFF and N/A state in 44.88 dB SPL.
2026-04-08T18:56:00.990157Z, Pump is now ON and Vacuuming state in 70.04 dB SPL.
2026-04-08T18:56:03.383201Z, Pump is now ON and Vacuuming state in 70.4 dB SPL.
2026-04-08T18:56:05.797165Z, Pump is now ON and Vacuuming state in 70.23 dB SPL.
2026-04-08T18:56:08.198243Z, Pump is now ON and Vacuuming state in 70.25 dB SPL.
2026-04-08T18:56:10.581880Z, Pump is now ON and Vacuuming state in 70.26 dB SPL.
2026-04-08T18:56:13.007419Z, Pump is now ON and Vacuuming state in 70.37 dB SPL.
2026-04-08T18:56:15.342795Z, Pump is now ON and Vacuuming state in 70.06 dB SPL.
2026-04-08T18:56:17.740343Z, Pump is now ON and Air leaking state in 70.51 dB SPL.
2026-04-08T18:56:20.154075Z, Pump is now ON and Air leaking state in 70.87 dB SPL.
2026-04-08T18:56:22.567857Z, Pump is now OFF and N/A state in 67.27 dB SPL.
2026-04-08T18:56:24.952904Z, Pump is now OFF and N/A state in 44.5 dB SPL.
2026-04-08T18:56:27.445644Z, Pump is now OFF and N/A state in 43.96 dB SPL.
2026-04-08T18:56:29.842277Z, Pump is now OFF and N/A state in 44.76 dB SPL.
2026-04-08T18:56:32.189130Z, Pump is now ON and Vacuuming state in 70.04 dB SPL.
2026-04-08T18:56:34.612787Z, Pump is now ON and Vacuuming state in 70.4 dB SPL.
2026-04-08T18:56:37.022362Z, Pump is now ON and Vacuuming state in 70.23 dB SPL.
2026-04-08T18:56:39.407554Z, Pump is now ON and Vacuuming state in 70.25 dB SPL.
2026-04-08T18:56:41.818739Z, Pump is now ON and Vacuuming state in 70.29 dB SPL.
2026-04-08T18:56:44.206867Z, Pump is now ON and Vacuuming state in 70.37 dB SPL.
2026-04-08T18:56:46.625545Z, Pump is now ON and Vacuuming state in 70.22 dB SPL.
2026-04-08T18:56:49.034446Z, Pump is now ON and Air leaking state in 70.48 dB SPL.
2026-04-08T18:56:51.522705Z, Pump is now ON and Air leaking state in 70.9 dB SPL.
2026-04-08T18:56:53.947800Z, Pump is now OFF and N/A state in 63.18 dB SPL.
2026-04-08T18:56:56.939722Z, Pump is now OFF and N/A state in 44.35 dB SPL.
2026-04-08T18:56:59.492322Z, Pump is now OFF and N/A state in 44.63 dB SPL.
2026-04-08T18:57:01.912682Z, Pump is now OFF and N/A state in 67.81 dB SPL.
2026-04-08T18:57:04.306329Z, Pump is now ON and Vacuuming state in 70.32 dB SPL.
2026-04-08T18:57:06.807582Z, Pump is now ON and Vacuuming state in 70.16 dB SPL.
2026-04-08T18:57:09.239453Z, Pump is now ON and Vacuuming state in 70.13 dB SPL.
```

- creationTime: 2023-03-16T05:09:48Z
- sender: eunseobkim
- instanceId: 1678943320
- version: 1.6.0.7
- bufferSize: 131072
- nextSequence: 13
- firstSequence: 1
- lastSequence: 12

Device: vacuum_pump; UUID: 001

Device : vacuum_pump

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2023-03-16T05:09:47.405461Z	SoundLevel		sound_level	spl	12	45.29

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2023-03-16T05:09:43.360865Z	Availability		avail	001	10	AVAILABLE
2023-03-16T05:09:43.360865Z	Execution		execution	e1	8	OFF
2023-03-16T05:08:40.188492Z	AssetChanged			vacuum_pump_asset_chg	4	UNAVAILABLE
2023-03-16T05:08:40.188492Z	AssetRemoved			vacuum_pump_asset_rem	5	UNAVAILABLE
2023-03-16T05:09:43.360865Z	State		vacuum_state	vs1	9	N/A

Figure 9 MTConnect agent '/current' response

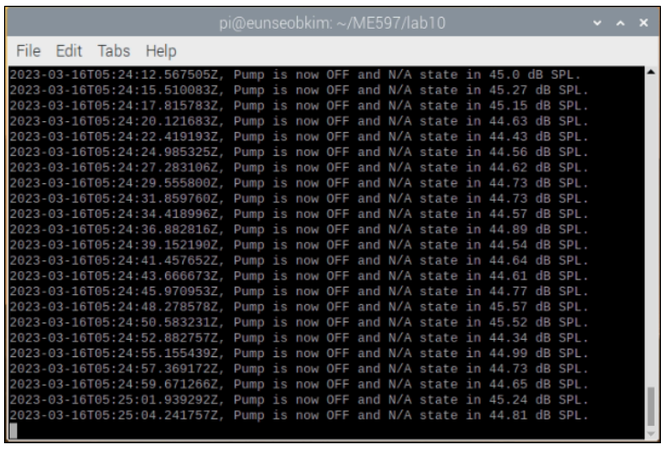


Figure 10 Capture of terminal in MTConnect adapter is running

3.2 MySQL database and Grafana dashboard

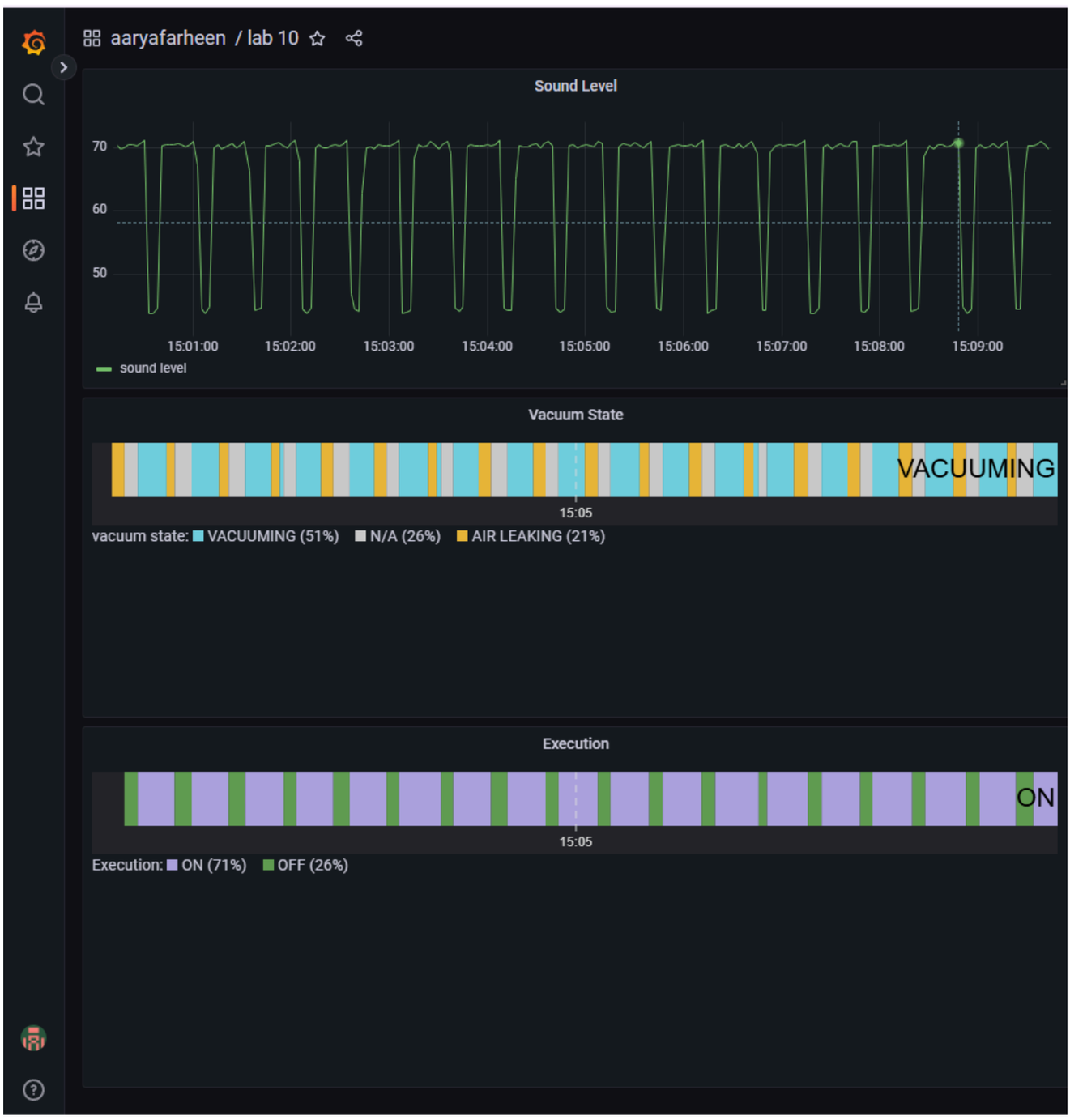
Let's move on to the data aggregation for MySQL database and visualization by Grafana. To collect MTConnect agent data, use the sample Python script (mtconnect_collecor.py).

MySQL and Grafana information is the same as the previous lab. However they are summarized again below.

- **MySQL**
- DNS: mepotrb16.ecn.purdue.edu
- Port: 3306
- Account: yourname (e.g., johndoe)
- Database (schema): ME597Fall24
- Table: yourname_lab6 (e.g., johndoe_lab6)
    - Reuse the table you created in Lab 6.
- **Grafana**
- DNS: mepotrb16.ecn.purdue.edu
- Port: 3000
- Grafana web page URL: http://mepotrb16.ecn.purdue.edu:3000/
- Account (username): yourname (e.g., johndoe) OR your purdue email account

The sample Grafana dashboard is 'Lab10 sample' in '0_SAMPLE' dashboard folder as Figure 11. Perform Task 3.2 to conclude this lab.

Figure 11 Lab10 Grafana dashboard sample



Task 3.3 Further discussion

Probably, sometimes you may see false results as Figure 12. In this example, although all actual state (red circle in Figure 12) was Air leaking, sporadically the inference state might be Vacuuming or vice versa. Discuss how to fix this problem in any aspect. For instance, we can discuss various aspects such as machine learning/deep learning architecture selection, training, algorithm, and so on.

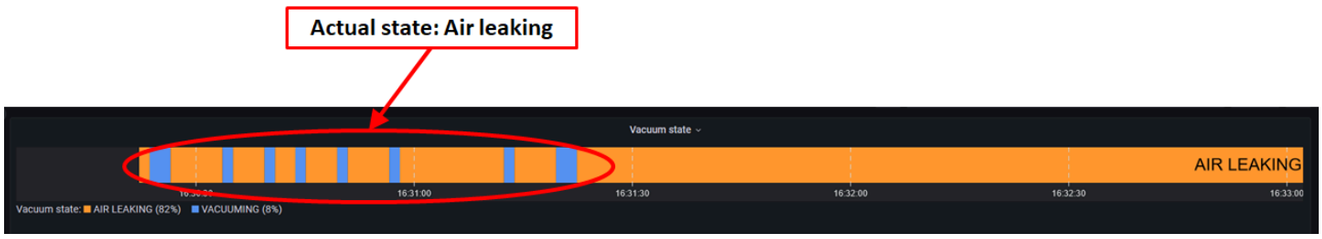


Figure 12 False prediction result

Possible Ways to Fix Sporadic False Predictions

- Use **prediction smoothing / majority voting** over several consecutive outputs instead of using only one instant prediction. This can reduce sudden label changes caused by noise.
- Improve the **training dataset quality and balance** by collecting more sound samples for each class, especially for confusing states such as *Air leaking* and *Vacuuming*. More diverse data can help the model generalize better.
- Refine the **feature extraction process**, such as testing different spectrogram settings, window sizes, overlap, or additional audio features, so that the difference between classes becomes clearer to the model.
- Try a more suitable **deep learning architecture** or tune hyperparameters. For example, adjusting the number of CNN layers, filter sizes, dropout, or using CNN-LSTM models may improve temporal understanding of sound patterns.
- Add a **confidence threshold or decision logic** so that predictions only change when the model is sufficiently confident for several consecutive time steps. This helps avoid rapid switching between classes.

Lab10 Summary and Deliverables

Answer the following questions for your achievements

Q1. Please summarize Lab10.

In this lab, we developed an intelligent system to monitor the operational state of a vacuum pump using sound signals. We trained a CNN-based machine learning model on Mel-spectrogram features extracted from audio data to classify different pump conditions such as OFF, ON-air leaking, ON-vacuuming, etc.

The system was deployed on a Raspberry Pi, where real-time sound data was streamed using MTConnect. The incoming audio signals were processed, converted into features, and fed into the

trained model for live inference. Based on the model's prediction, the system identified the current operational state of the pump.

Additionally, we integrated the system with MTConnect Adapter and Agent, enabling structured data streaming. The output was visualized using Grafana, allowing real-time monitoring and analysis of the pump's condition through a dashboard.

Overall, this lab demonstrated the integration of machine learning, embedded systems, industrial communication (MTConnect), and data visualization, forming a complete real-time condition monitoring solution.

Q2. What skills did you have to develop to accomplish this project?

- Applied **Machine Learning (CNN)** for audio-based classification
- Worked with **audio signal processing** (RMS, FFT, frequency analysis)
- Integrated **MTConnect (Adapter + Agent)** for industrial data streaming
- Built end-to-end pipeline: **Sensor → ML Model → Output → Visualization**
- Visualized real-time data using **Grafana dashboards**

Q3. What aspects of this project were the most beneficial for your learning?

🎯 Most Beneficial Aspects of This Project

- Learned how to connect **signal processing + machine learning + embedded systems**
- Understood challenges of **real-time data processing and inference**
- Worked with **industrial protocols (MTConnect)** used in smart manufacturing
- Developed ability to **visualize and interpret live data using Grafana**
- Built confidence in handling **multi-component engineering systems**

Q4. What challenges did you encounter in completing the project?

⚠️ Challenges Faced

- Debugging **model deployment issues on Raspberry Pi** (library compatibility, performance)
- Resolving **TensorFlow/Keras model loading and format errors (.h5 vs SavedModel)**